Thesis Title

Development of Deep Learning-based Automated System for Mango Leaf Disease Diagnosis and Classification

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MSIT-22-03

2022-2024

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Thesis submitted to the Department of Information Technology

Bahauddin Zakariya University, Multan

In partial fulfillment of Requirements of MS Degree

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BAHAUDDIN ZAKARIYA UNIVERSITY MULTAN

PAKISTAN

FINAL APPROVAL

It is certified that content of thesis entitled "Development of deep- learning based automated system for Mango leaf disease diagnosis and classification" have been submitted in the partial fulfillment of the requirement of the degree MS (IT) by **Sana Saeed** and this thesis accepted by Bahauddin Zakariya University, Multan.

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DEDICATION

I want to dedicate this thesis to my beloved Mother who encouraged me to pursue my MS degree. Even when I doubted myself, her trust, prayers and support remained a constant source of inspiration, driving me to reach this milestone. This work is the testament to the endless love and inspiration you have given me. I am forever grateful for her love, guidance, and constant support. I hope this will not be my last academic achievement; there will be many more I can dedicate to you.

ACKNOWLEDGEMENT

First and foremost, I am extremely thankful to Allah Almighty for giving me strength and courage to complete this challenging task. I would like to express my deepest gratitude to my supervisor, **Dr. Maruf Pasha** for their invaluable guidance, support, and encouragement throughout the course of my research. Their expertise, patience, and insightful feedback have been instrumental in shaping this thesis and helping me overcome the many challenges I faced along the way.

Thank you for your constant belief in my potential and for pushing me to achieve more than I thought possible. Your commitment to my academic and personal growth has left a lasting impact, and I am sincerely grateful for the privilege of working under your supervision. This thesis would not have been possible without your unwavering support.

ABSTRACT

Deep learning is making a lot of progress in every field so as in agriculture sector. For disease detection in plants, deep learning models promised to provide more accuracy, robustness and generalization in models. Deep learning models provide cost effective, faster solutions for detection of disease for farmers and agriculture experts to accurately identify the plant diseases. In this thesis, automated deep-learning based system developed for mango leaf disease diagnosis and classification. For this research, dataset has been acquired from online database (Kaggle) that contain total eight classes including seven disease classes and one healthy category. Three deep learning models are used that are CNN, Squeeze Net, and Alex Net. The system is evaluated by testing the dataset, presenting its effectiveness in detection of mango leaf diseases. Squeeze Net turned out to be the best model for disease diagnosis and classification. This model offers highest accuracy of 99.6% that is greater than other two models CNN, and Alex Net. Squeeze Net also achieved more precision and recall as compared to other two models. This research aims to contribute in precision agriculture and help farmers to timely diagnosis the diseases to reduce the production loss of the crops.

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INTRODUCTION

The scientific name for mango is “Mangifera indica” which includes high nutritional and economic value around the globe. Multiple diseases affect the quality and quantity of mangoes and causing an impact on the overall production of this fruit (Faye, Diop et al. 2023). The diseases in plants are natural and detection of diseases is significantly important in agriculture (Wagle and R 2021). Agriculture provides a lot of help in dealing with food insecurities if diseases on plants can be identified on time. Every year almost forty percent of the crops worldwide are affected by plant diseases. There are many types of diseases in plants especially those that infect leaves including fungus, viruses, and bacteria that reduce overall agricultural productivity if not treated properly. Early detection of diseases in plants may result in increased crop production

There are different types of leaf diseases affecting mangoes productivity but in this research, our main focus is on seven diseases namely anthracnose, powdery mildew, bacterial canker, sooty mould, gall midge, cutting weevil, and die back.

1.1 Overview of mango leaf diseases

There are variety of diseases including bacterial and fungal diseases that can affect the healthy and productivity of mango crops. Following seven diseases of mango leaves are most common types of diseases. These diseases cause huge loss in production of mangoes so early detection and proper remedies can help a lot.

Anthracnose: Anthracnose is basically a fungus that creates dark patches on leaves. After that leaves become yellow and brown and cause leaf loss. The dark spots on the leaves are the initial symptoms of this disease leading toward black patches on flowers and fruit. The main causes of this disease are wind and rain and the use of fungicides helps to cure this disease .

Powdery mildew: This disease is also a fungal disease that deformed leaves and they fall off the tree. White and powdery material started appearing on leaves by this disease. This disease spread by wind and insects.

Bacterial canker: This disease causes the branches, leaves, and stalks to become dripping and then turn into canker.

Sooty mould: This disease is caused by honeydew (sticky secretion part that attracts other insects) that is used as food by insects, mould disease spreads overall on the leaf and turns black.

Gall midge: This disease causes pimples on leaves and leaves start falling from the plant resulting in reduced crop yield.

Cutting weevil: This disease is really harmful because it cuts the leaves in such a way that it looks like cutting by scissors impacts mango yield.

Dieback: This disease causes the mango branches to dry out turn brown and fall off the tree (Ahmed, Ibrahim et al. 2023).

Beside these seven disease categories, healthy leaves are also included in dataset and there are total eight classes for disease diagnosis and classification in this research.

Further in this chapter, there is an introduction to deep learning, Fundamentals and Applications of Deep learning, and the role of deep learning in agriculture.

1.2 Introduction to Deep Learning (DL)

Deep learning is a branch of machine learning that has revolutionized various domains like computer vision, natural language processing, voice recognition, healthcare, and many more fields. A huge development has seen in machine learning and Artificial Intelligence in the past few years (K Dokic, L Blaskovic et al. 2020). But, deep learning has remained unpopular because it went through many different names. The history of deep learning is very long and rich. The development in deep learning was named cybernetics from 1940 to 1960 and then became popular with the name of connectionism from 1980 to 1990 in the third phase of research and development in 2006 it became deep learning reflecting influence on many aspects of research and development.

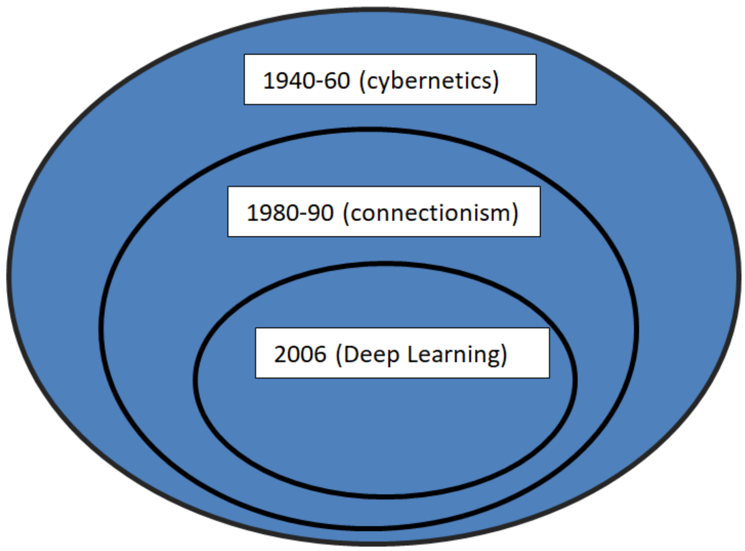

Figure 1: Evolution of Deep Learning

The earliest algorithms in deep learning were computational models developed based on biological learning like how learning happened in brains. Artificial Neural Networks (ANNs) considered as most popular model in deep learning.

Neural Networks had also been used by machine learning but these models did not realistically work as biological learning but deep learning models were engineered inspired by the human brains and connected with network of neurons.

The modern term "Deep Learning" goes beyond this neuroscientific purpose and nowadays it uses more appealing principles of learning and is not only inspired by neural networks. With the advent of technology deep learning has become successful in commercial applications as well as in art and technology and handles large and complex amounts of data. A few important features of deep learning play an important role in its success:

- **Increased sizes of datasets:**

Deep learning models required skills to get good performance from its algorithms but the main reason for the success of deep learning that it became more useful when the amount of training data has increased. The size of dataset was a benchmark and over the time it has been changed and now deep learning algorithms and models performs really well with larger datasets. In this age of “Big data”, deep learning handles large amount of unlabeled data working with supervised and semi supervised learning.

- **Increased model size:**

Another important reason for the success of deep learning is increased model size.

Model size has been increased due to high speed computers with larger memory and due to the availability of larger datasets. Deep learning models have grown much more in sizes due to improved hardware and software infrastructure of deep learning.

Besides that deep learning has solved very complicated applications with high accuracy over time. Further advances in deep learning depend on the advancement of hardware and software infrastructure. Deep learning achieved huge success in image segmentation and in pattern and voice recognition over time. Deep learning models can perform better because several connections between artificial neuronal networks depend on hardware capabilities (Goodfellow, Bengio et al.).

Traditional machine learning approaches require manual feature extraction but on the other side deep learning automatically learn and extract features through multiple abstraction layers.

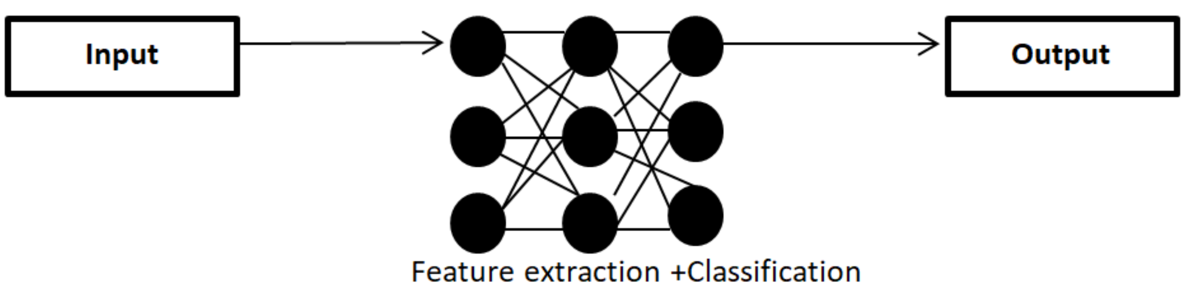

Figure 2: Working of Deep Learning Models

1.3 Fundamentals of Deep Learning

The core of deep learning is neural networks. Deep learning is a part of Machine learning and Convolutional Neural Networks belong to deep learning. The difference between artificial neural network and deep learning networks are the multiple hidden layers present between input and output layer. The main layer is convolutional layer in deep learning models. There are different types of convolutional models in deep learning such as AlexNet, SqueezeNet, GoogLeNet, ResNet, VGG16, and DenseNet. The only differences between these models are number of layers and nonlinear functions used in these models. Otherwise, all of these models have similar structure like four layers including convolutional layer, output layer, max pooling layer, and fully connected layer(Wagle and R 2021).

The number of connections between the neurons in the artificial neural networks was limited in the beginning because of limited hardware capabilities. Nowadays the number of connections between neurons is increased and also design-specific(Goodfellow, Bengio et al.).

1.4 Applications of Deep Learning

The huge success of deep learning has led it towards widespread use of it in various applications.

1.4.1 Computer vision

Computer visions has considered as one of the most traditional research field in deep learning. The most important task for computer vision in deep learning includes object recognition and optical character recognition became very popular.

There has been wide use of computer vision in image segmentation, object detection, and in facial recognition.

1.4.2 Natural Language Processing (NLP)

Deep learning models revolutionized NLP by generating semantic and performing sentimental analysis.

1.4.3 Healthcare

Deep learning has widely used in healthcare in disease diagnosis, drug delivery, and in critical treatment planning.

1.4.4 Agriculture

Deep learning has making many efforts in agriculture for disease diagnosis and classification as well as in control, management, and enhancement of crop production (Singh, Chouhan et al.

2019). Automated systems in deep learning achieved high accuracy in diagnosis and classification of diseases(Kumar, Ashtekar et al. 2021).

1.5 Role of Deep learning in agriculture and disease diagnosis

Agriculture is the backbone of the economy in various countries around the world. 70% of the world economy depends on agriculture. The huge increase in population raised the demand for high quantity and quality of food in the world. However various diseases affect the production and yield of the crops (Sharma, Suvarna et al. 2022). Traditional methods of disease detection in agriculture depend on experts like plant pathologists, who inspect diseases visually. This method was time-consuming and costly. In previous years, deep learning has acted as a promising tool that automates the diagnosis and classification of diseases in plants.

The relationship between deep learning and agriculture plays a significant role in better production of crops and early disease detection. Before deep learning, machine learning is making remarkable progress in precision agriculture and disease detection.

There are several benefits that deep learning provides in agriculture:

1.5.1 High accuracy

Deep learning models especially Convolutional Neural Networks (CNNs) have achieved high accuracy in the diagnosis and classification of diseases. These models can learn complex features and patterns of specific diseases and provide better results and diagnostic accuracy.

1.5.2 Speed and Efficiency

Deep learning models perform processing on data in less time, train the data, and enable quick and efficient disease classification. In the large-scale-agriculture setup, speedy diagnosis of diseases is important in critical disease management.

1.5.3 Crop monitoring and management

Deep learning models can detect diseases and weeds in crops and provide high accuracy by analyzing the image datasets. The early diagnosis helps farmers in doing remedies and pesticides for the proper management of diseases.

1.5.4 Cost-effectiveness

Choosing automated disease diagnosis with the help of technology like deep learning can provide long-term benefits to farmers. The initial setup requires resources but these deep learning models have the potential for early detection of diseases and lead toward significant savings by reducing the crop losses.

1.5.5 Precision Agriculture

Precision agriculture is the modern technology of deep learning that offers refined techniques for optimizing the crop production. Economic development in agriculture can be achieved by making use of these technologies.

But there are some challenges faced by deep learning in agriculture:

1.5.6 Data quality and quantity

The major challenge faced by deep learning models is that they require large volume of data with high quality for training. In agriculture, it is a challenging task to clean and label data due to variability in environmental conditions. Obtaining image data with high quality is also another challenging task with accurate annotation, particularly for novel diseases.

1.6 Problem definition

In this research, the problem statement revolves around the development of an advanced mango leaf disease detection system that leverages deep learning techniques to enhance the performance of existing algorithms and optimizing precision and recall. The main objective of this research is to develop a solution that is capable of accurately classifying different types of mango leaf diseases while overcoming the limitations of associated individual algorithms in this research.

1.7 The motivation of Research

Technology plays vital role in every field of life so as agriculture. There are many methods available for disease diagnosis and classification. In past few years, a lot of work has done in field of agriculture by using machine learning and deep learning models. But considering the literature there is still no work on diagnosis of mango leaf diseases that covers multiple categories of diseases. So, this work motivates to classify seven categories of diseases including Anthracnose, Bacterial canker, Cutting Weevil, Die back, Gall midge, Sooty mould, and

Powdery mildew and develop an automated system using different Convolution Neural Networks (CNNs) and training datasets two optimization algorithms.

1.8 Outline of thesis

Thesis is organized in 6 chapters. Literature Review is discussed in chapter 2. Chapter 3 demonstrates problem statement; proposed methodology is presented in chapter 4. Experiments and Results are shown in chapter 5 and chapter 6 include comparison of models and 7 chapter discuss conclusion and future direction of the thesis.

Chapter 2

BACKGROUND STUDY

2.1 Introduction to Background study

This chapter presents the background study of mango leaves disease diagnosis using deep and machine learning techniques. In past few years, there have been a lot of researches on plant leaf disease detection using machine learning models. Accurate crop protection and automated disease diagnosis depend heavily on machine leaning techniques. The potential of deep learning to transform automated systems for diagnosis of mango leaf diseases have attracted a lot of attention recently. This chapter reviews the literature that has been presented about mango leaf diseases, with an emphasis on the methods and results of those works.

2.2 Background study on image processing techniques

For the development of an automated system several kinds of image processing methods are essential for classifying and detecting mango leaf disease. (Trang, TonThat et al. 2019) applied deep neural network for detection of diseases in mango leaves. The image processing techniques involved rescaling and center alignment. The first one is rescaling and then followed by center alignment. In the rescaling phase, images have been converted to lower resolution. In a dataset, there are many images with lower visual quality, which would not help in extracting features from some regions in images. Considering a leaf region includes a variety of contrasts, contrast enhancement helped to change the pixel intensities and providing more insight about certain region of disease leaves.

(Sanath Rao, Swathi et al. 2021) presented transfer learning technique for disease detection of mango and grapes leaves. Image processing, image acquisition and image segmentation has performed before classification. Images has been captured using high resolution camera and resized to 256× 256 pixels, converted into RGB images. In this study, background of the images has also segmented and this step operates very well. (Jain and Jaidka 2023) proposed hybrid model for diagnosis of mango leaf diseases. The machine learning algorithm SVM and Stochastic Gradient Descent (SGD) have been implemented in this work and does not discussed image processing techniques in detail. (Manoharan, Thomas et al. 2021) performed data augmentation and trimming of images on dataset as preprocessing step.

The approach of (Selvakumar and Ananthakrishnan 2022) for image processing include resizing images to 256×256 pixels helps to achieve high accuracy and performed image augmentation like image rotation, affine transformation and strength transformation. These preprocessing techniques turn out to be beneficial in feature extraction as well as in reducing training time.

(Saleem, Hussain Shah et al. 2021) presented CNN for classification of mango leaf diseases. This study included the image preprocessing in two steps: 1. LCHR contrast stretch 2. RGB conversion. Besides these techniques Resizing and image augmentation has also been performed on dataset. Local-Contrast-Haze-Reduction (LCHR) approach helps in reducing errors that affected system's performance and reduce the execution time with increase in accuracy.

The idea behind this contrast stretch has to improve the low contrast areas of diseased images.

These preprocessing steps have an effective outcome resulting as noise removal, background correction, and highlighting the healthy and diseased image regions.

A combination of data augmentation techniques has proposed by (Faye, Diop et al. 2023) for identification of diseased mango leaves. The main goal of this study is to implement several data augmentation methods, including affine transformation, noise injection, blur, contrast and brightness, zoom, and image flipping. Noise injections helps to enhance the brightness and color of images and blur technique made images less sharp. The contrast has been adjusted to 0.5;2;2.5 in this study and images has flipped in vertical as well as in horizontal direction. At the end, affine transformation applied on whole dataset combining translation, rotations, and shear dilations. Affine transformation proved as very helpful because images has been captured from different camera positions and projections.

2.3 Background study on Deep and Machine learning Algorithms

(Trang, TonThat et al. 2019) presented deep neural networks techniques to diagnose diseases of mango leaves. The dataset for this research has been taken from an Giang province of Vietnam which includes 394 images from four classes: healthy leaf images, gall midge, powdery mildew ailments, and anthracnose images. The dataset's images have been captured in 3096×3096 pixels resolution with no background. The proposed method for the identification of disease consists of four stages, image preprocessing, resizing, center alignment, contrast enhancement, and classification. Convolution neural networks (CNNs) were then utilized to classify the input images. The suggested framework residual network obtained 88.46% accuracy which was greater as compared to other pre-trained models on mango disease dataset namely Inception v3, Alexnet, and Mobilenet v2.

(Sanath Rao, Swathi et al. 2021) presented neural network for identification of grapes and mango leaf diseases. The dataset contained total 8,438 images including both diseases and healthy leaf images of mango and grape leaves. The dataset used in this study was gathered locally and includes 1,216 images from the plant village and mango leaf datasets. Alexnet, a pre-trained CNN architecture, has been utilized for feature extraction and classification. An app called "JIT CROPFIX" was created on Android benefits farmers to implement disease detection systems on their smartphones. The proposed model achieved 89% accuracy in mango leaf disease detection and 99% accuracy obtained for grapes leaf diseases.

(Jain and Jaidka 2023) presented a hybrid method that combines stochastic gradient descent (SGD) and SVM techniques for diagnosis of leaf diseases. In this research, the fundamental Harumanis mango leaves 2021 dataset—which includes 1,206 images of healthy and diseased subjects—was obtained from Kaggle. The suggested model used a hybrid strategy of SVM and SGD for classification after first customizing SVM to extract the deep characteristics of the images. Results showed that this model achieved 97.7% accuracy in classification of mango leaf diseases.

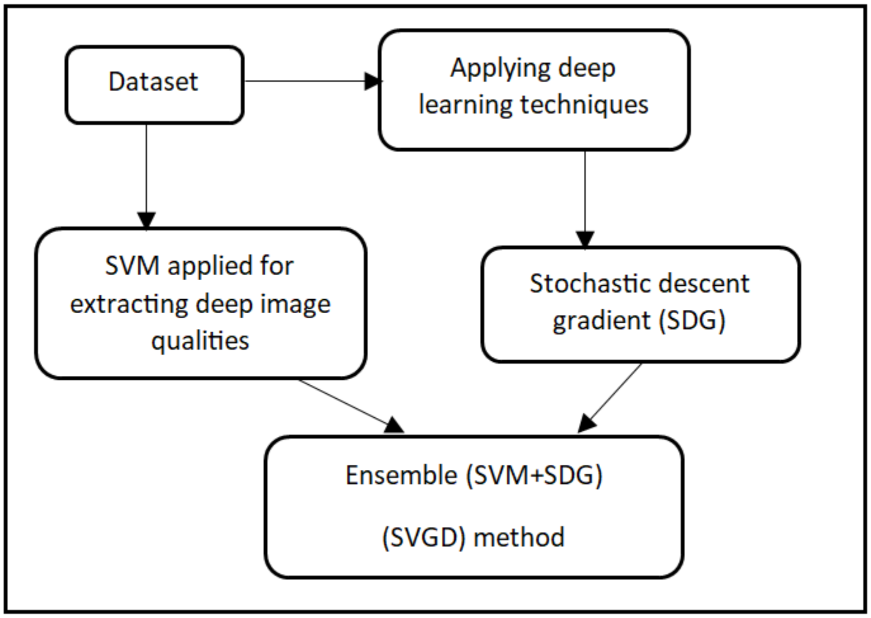

Figure 3: Existing methodology of Mango leaf diseases detection

Another study using deep learning algorithms for classification of mango leaf diseases by (Manoharan, Thomas et al. 2021) presented a customized CNN algorithm which demonstrate statistically improved accuracy of 98%. In this study, data augmentation and resizing of input images has done in preprocessing step. The next step carried out as CNN model training .The proposed CNN model was compared with AlexNet and VGG16 that obtained 61% and 62% accuracy accordingly.

(Rahaman, et al. 2023) presented a smartphone application based on deep learning which can identify diseases in mango leaves and suggest appropriate pesticides. An Android application has been developed that integrates machine learning techniques including DenseNet169 to identify mango illness and provide a recommendation of pesticides after disease detection. For disease detection, the dataset has been taken from 20 different sources and normalized these samples by removing noise from the images. There are different deep-learning techniques were used on the dataset including Inception V3, MobileNet V2, AlexNet, and VGG16. These models were tested and results showed that DenseNet169 achieved the maximum accuracy of 97.81%.

The findings of (P. and K. 2023) presented in their study a hybrid model aimed at revolutionizing mango leaf disease detection. There were seven types of fungal and other diseases that infect mango leaves discussed in this research. The dataset for this study has been taken from mendley and from some local farms in India consisting of total 4873 images. For feature extraction from the input images, Fast Fourier Fransform (FFT) technique was used that's really helpful for identification of disease. The proposed hybrid model ensemble the customized CNN with EfficientNetB4 and achieved 93.01% accuracy in detection of diseases.

(Selvakumar and Ananthakrishnan 2022) presented in their study an ensemble technique stack, of logistic regression algorithm, support vector machine and well known neural network for diagnosis of mango leaf. In this study, existing methodologies of these four algorithms have been studied and then implementing stack technique that achieved higher accuracy in classification of mango leaf diseases as compared to other algorithms. The input images were preprocessed and then classification will be performed. In pre-processing, image resizing, rescaling and arrangement conversion was done. The obtained accuracy of this finding was 98.6% and can be further improved by Adam optimization to 99.2%.

(Saleem, Shah et al. 2021) adopted novel vein pattern and segmentation technique for classifying sick mango leaves. The proposed approach for this study was leaf vein segmentation, which extracted the image's vein pattern. In extant literature, there were two diseases termed powdery mildew and sooty mold that produce an additional layer on the leaf and disturb the leaf vein pattern. This study proposed a solution for extracting the leaf pattern. Image resizing and data augmentation has performed for preprocessing of the input images. For feature extraction and fusion, canonical correlated analysis (CCA) based fusion has been performed. For calculating the classification accuracy, Ten different kinds of classification algorithms were used.This study proposed a unique technique of segmentation that have not been applied earlier in mango leaves disease detection.

(S. Arivazhagan and Ligi 2018) presented CNN model for diagnosis of mango leaf diseases. There were five diseases including anthracnose fungal disease, alternaria leaf spot on mango leaves, leaf gall, leaf webber, and burn on leaves. The dataset consisted of 1200 input images of both healthy and diseased leaves. The ReLU Layer, Max-Pooling Layer, Dropout Layer, Batch Normalization Layer, and hidden layer worked between input layer and output layer of the model. The study's findings demonstrated that proposed model obtained 96.67% accuracy by applying CNN for identification of diseased mango leaves.

(Mehta, Kukreja et al. 2023) findings presented Federate learning (FL-CNN) model for disease detection of mango leaves and severity analysis. In this research, there were total four severity categories including beginning, mild, moderate and severe. The proposed FL-CNN model trained locally by distributing data among various clients and also considering the factor of data privacy. For performance measurement, macro average, micro average and weighted average parameters had used. There were total four clients and first client achieved lowest macro average and weighted average 93.92% and 93.96% accordingly, and the fourth client achieved highest macro average and weighted average of 96.11% and 96.09% respectively. The FL-CNN model provided the benefit of data privacy, lower communication costs, and improved performance in classification of mango leaf diseases.

(Faye, Diop et al. 2023) presented a research that diagnosed diseased mango leaves using data augmentation methods including blur, zooming images, contrast enhancement, and affine transformation. The results of each of these techniques on ResNet50 model has been observed on small dataset. Four diseases discussed in this paper were anthracnose, gall midge, sooty mold, and powdery mildew. Results showed that contrast & flip & affine transformation technique was the most effective combination for classifying diseases of mango leaves, obtaining a f1-score>0.9, testing accuracy of 97.80%, and training accuracy of 98.54%.

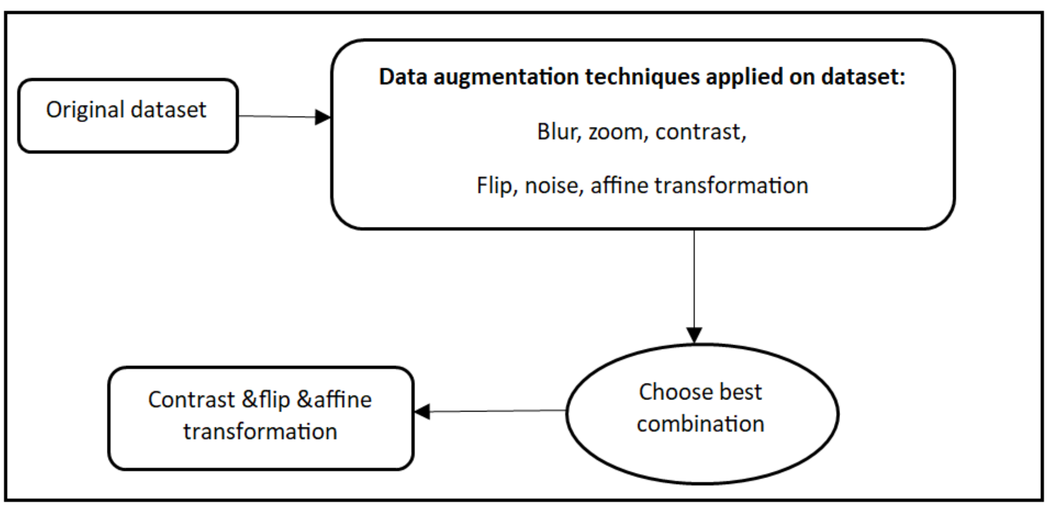

Figure 4: Existing Methodology using data augmentation techniques

(Singh, Chouhan et al. 2019) applied Multi- layer CNN (MCNN) for detection of the disease anthracnose that affects mangoes. The primary goal of MCNN model was to improve computer vision of deep learning techniques for precise disease detection. The dataset for this study was self-acquired and also taken from plant village. The central square approaches for image scaling and histogram equalization for contrast enhancement were the preprocessing techniques employed in this study. The proposed MCNN model obtained 97.13% accuracy and provides cost effective solution in contrast to alternative cutting-edge methods. Accurate classification of mango leaf disease was made possible by the MCNN model's lightweight and efficient computation.

Another study by (Mia, Roy et al. 2020) presented ensemble neural network for classification of mango leaf diseases. In this study, classification technique was used to train data for recognition of mango leaf disease by collecting infected images. The machine learning ensemble approach automatically identifies the diseased leaves by uploading new images and comparing them with trained data. After comparing both trained images with uploaded images, diseased images were classified. Results showed that the proposed system achieved 80% in the classification of mango leaf diseases. This study showed support vector machines (SVM) and neural networks (NN) ensemble for detecting leaf diseases offer higher accuracy.

(Venkatesh, N et al. 2020) presented convolution neural network model using transfer learning for the classification of mango disease known as Anthracnose. The VGGNET model used in the present study has been improved to incorporate the best features of the Inception module with VGGNET for detecting the diseases on mango leaves. The basic characteristics were extracted with the help of the VGGNET module; nonetheless, Inception module performed image classification. In this paper, for the classification of the image features comprised the following: leaf margin, leaf shape, petiole condition, leaf color, venation, and tip form. The dataset contained 2268 images where 1198 were self-acquired images and 1070 images were taken from plant village. The proposed transfer learning V2IncepNet model achieved 92% in classification of diseases. The results of the research demonstrated that the suggested approach for classifying diseases of mango leaves was simple yet effective.

(Math and Dharwadkar 2022) findings presented using convolutional neural networks, grape diseases may be detected and diagnosed early. The study's findings suggested CNN design resembles to the traditional CNN architecture, with the exception of the flattening layer's replacement of the global average pooling (GAP). Using transfer learning, a proposed deep learning convolutional neural network (DCNN) was developed to deliver more accuracy than pre-trained models. This study presented five models including VGG16 and 19, ResNet 50, V3 inception and last one Xception model were trained before being compared to a customized model. The study's findings demonstrated that, when using RGB leaf images to identify grape diseases, the model's accuracy was 99.34%. Real-time mobile devices usage of the suggested model in TensorFlow tfilte format is feasible for identification of grape leaf diseases on early stages and provided precision agriculture (PA) application.

Hybrid approach for detection of mango leaf detection presented by (Rajpoot, Dubey et al. 2022) that combined convolutional neural network (CNN) Bi histogram equalization for preserving the brightness (BBHE). BBHE and CNN were commonly used in disease detection system for feature extraction. Preprocessing of input images were done by flattened, scaled down, and finally converted to their threshold values. After preprocessing, relevant features were extracted from the images for the purpose of classifying the images, the suggested hybrid model used computationally limited resources. In this study, there were total four algorithms discussed BBHE algorithm, canny edge detection, hybrid BBHE and CNN algorithm, and selective retinex fusion are a few applications. Results of the proposed models showed that hybrid model achieved highest accuracy of 95.65% and performed better as compared to other approaches. Another study proposed by (Kumar, Ashtekar et al. 2021) proposed disease detection infected by fungal disease Anthracnose using deep convolutional neural networks (CNNs). The dataset for this study has been captured from real environment and second category of images taken from Plant Village. Preprocessing of images involved four steps included cropping, histogram equalization for contrast enhancement, data augmentation that consist of rescaling, shear and zoom, and flip. The proposed methodology obtained 96.16% accuracy for classification of Anthracnose disease in mango leaves.

For grape disease detection Mawaddah Harahap et al presented a study using dual channel convolutional neural network (DCNN). The Gabor filter approach was employed on input images for feature extraction. The segmentation-based fractal co-occurrence texture analysis approach has also been used to extract characteristics from leaves, such as color and texture. The model's execution procedure took longer when the dataset was larger. The research was limited in that the efficiency was lowered throughout testing because the Gabor method's angle and frequency parameters changed. The DCNN model's accuracy was 91.67%, however the angle and frequency variables need to match those used during training. The suggested model can identify disease more quickly and effectively if the angle and frequency data are provided precisely as they were during training and testing.

(Aditi Singh and Kaur 2021) findings presented a research for detection of potato plant leaf diseases using machine learning algorithms. Two different types of diseases including early and late blight presented in this research. To segment images, machine learning methodology of K-means was used in this study. For segmentation of input images, features using the gray level co-occurrence matrix were extracted. The suggested approach classified potato plant diseased leaves using SVM method. In this research, proposed framework achieved 95.99% accuracy in classification of diseased leaves. For improvement in accuracy, author suggested using artificial neural networks, particular convolutional neural network for future work because CNN outperformed existing modern methods by achieving higher accuracy.

(Jackulin and Murugavalli 2022) presented a review study for detection of plant leaf diseases using machine and deep learning. There were several approaches for segmenting plant diseases, such as the threshold method, clustering, regional and edge detection method. Further in this study, advantages and disadvantages of these techniques were discussed that threshold method was fast, simple and very low cost method computationally. But disadvantage showed that this method was extremely sensitive to noise. Second was clustering method that showed that homogenous regions were easy to obtain but clustering had worst case scenarios when this method did not work well. Edge detection method worked well with systems where images have high contrast regions but this method did not work well with edgy pictures. At last the regional method provided better performance as compared to other approaches but required more memory and computational power. This research presented a comparative study between machine learning and deep learning algorithms. When it comes to machine learning techniques, there were most commonly used techniques included Naïve Bayes (NB) classifier, KNN classifier, RF classifier, DT classifier and SVM classifier were discussed. For deep learning techniques CNN, Inception- V4, VGG-19 and 16 were discussed. All of these techniques were summarized for identification of plant leaf diseases. This review study helped in order to perform classification on diseased images dataset using any above machine learning or deep learning techniques.

(Prabu and Chelliah 2022) introduced a novel, improved approach for classifying leaf diseases using CNN architecture following a crossover-based Ivey flight distribution algorithm (CROLFD). Already there were many machine learning algorithms that solved the problem of disease detection but all of these models suffered from few limitations like computational complexity, overfitting problem and average performance due to higher feature dimensionality. The presented framework solved all of these problem and divided divided into four phases: (1) preparation of the data; (2) selection of features; (3) stage of learning and classification; (4) stage of evaluation of performance. Three diseases—bacterial black spot, sooty mold, and anthracnose—were covered in this study. The 380 images that make up the dataset for this study were self-acquired from Andhra Pradesh, India. Data augmentation methods including fill mode, flip horizontally and vertically, feature-wise standard deviation normalization, brightness, zooming, and shearing technique implemented to the input image, and rotation were used in a preprocessing stage to address the overfitting issue. A CNN-based crossover-based Ivey flight distribution technique was used to improve feature selection. MobileNet V2 was used as pre trained model for the learning and stage, and at the end of MobileNet V2, support vector machines (SVM) were employed to classify diseases. The model results suggested that, CROLFD had the greatest accuracy in classifying mango leaf diseases, at 94.5%. When the suggested CROLFD model's classification results are contrasted with those of other latest approaches, including Neural Network (NN) using SVM, which attained 72.63% accuracy, CNN achieved 84.13% accuracy Artificial neural network (ANN) achieved 80.48% accuracy.

(BALASUBRAMANYAMMEDA 2021) proposed a multi-class classifier Feedforward and Neural Network model for identification of mango leaves in early stage. The model presented in this study classifies leaf diseases by RGB value percent of the affected region by using image preprocessing techniques. In this study, two techniques were proposed (1) convolutional neural transfer learning (TF)-enhanced neural network (CNN) models and (2) feature-selected artificial neural networks (ANNs). The more than 450 images in the collection were self-acquired. This study addressed three diseases in total: powdery mildew, gall midge, and anthracnose. The images were captured in different lighting condition such as indoor (labs) or in fields.

Preprocessing of the input images included rescaling, and lowering the resolution of images as compared to original ones and the next step consisted of center alignment. The dataset of input images consisted of three stages: testing, validation, and training. From the total data 20 percent data set for testing data, and the rest of data was divided into two parts containing 80% data for training of model and 20% data required for validation of model. For feature selection, Contrast-Limited adaptive histogram equalization (CLAHE) and grey wolf optimization was used. This study developed a model that obtained accuracy 89.71% and 84.88% accordingly provided the benefits of the smaller model that performed with high speed.

(Nagaraju, Venkatesh et al. 2020) presented a customized model for transfer learning and help in classification of grape leaf and apple diseases. This fine-tuned network was VGG16 that classified eight different types of diseases of both diseases of grape and apple leaves. In this model, VGG16, existing last layer was removed and an output layer would be added. The dataset of this study was taken from publicly available dataset containing 17638 images out of which 8806 were apple images and 8832 images were grapes dataset. Transfer learning provided benefits by reducing the training time of the model by training only last layer of the VGG16 network. Preprocessing in the images were done in batch wise manner to remove the background noise and to smooth image edges. Experimental results showed that VGG16 model achieved 98.9% accuracy in training phase where in testing phase classification accuracy obtained was 97.87%.

(Saberi Anari 2022) suggested classification method using hybrid model for leaf diseases based on an ensemble technique and modified deep learning algorithm for agricultural AIoT-based monitoring. The hybrid model combined modified deep transfer learning methods, a feature extraction deep convolutional neural network. In this research, model engineering (ME) contributed to effective feature extraction. There were multiple models of support vector machines (SVMs), which are utilized to discriminate features, increasing processing speed, and improving the classification process. The dataset for this study has been taken from PlantVillage and UCI containing 90,000 images of corn, apple, cotton, rice, grape, and pepper. There were 3 types of diseases on apple leaves including scab, black rot, and cedar apple rust discussed in this study. Two types of diseases on corn leaf such as Northern leaf blight, common rust, black rot, and black measles are the three diseases that affect grape leaves; bacterial spot is the only disease that affects pepper leaves.

The different types of rice diseases including bacterial leaf blight diseases on leaves, brown spot on infected leaves, and leaf smut, were covered in this review, and four kinds of diseases on cotton leaves as Areolate mildew, Mela (soreshin), Myrothecium leaf spot, and Fusarium wilt. In the training step, radial basis functions (RBF) kernel parameters were determined for the selected model. Convolutional neural network (CNN) used to train models including VGG model, DenseNet, and ResNet model, and the ensemble learning of SVM models as final classifier helped in efficient classification of diseases.Results showed that deep CNN based dilated ResNet-18 model achieved 99.1% accuracy in classification of diseases and reduce computational complexity of the algorithm that helped to increase the execution time.

(Sema, Yayeh et al. 2023) findings suggested using the Histogram Oriented Gradient (HOG) method and CNN for self-sustaining identification of mango leaf diseases and classification. The 400 images that made up the dataset for this study came from the Research Institute Assosa Brach in Ethiopia. Other images were gathered from internet sources of mangoes affected by diseases such bacterial canker, powdery mildew, anthracnose, and algal spot diseases. Data preprocessing step helped to remove noise from images that were blur and had bad quality. All images were resized to 220×220 pixels led to a reduction in both the computing cost and image processing time. Histogram equalization techniques were used in preprocessing. To eliminate noise from image edges anisotropic diffusion filter has been used that reduced image edges while maintaining its smoothness. In image segmentation, threshold segmentation technique used that was easy to implement and computationally fast. Features from the images were extracted using CNN for training and testing the model. For extracting both texture and shape from images and to perform the extraction process more effectively, CNN and HOG were both employed as hybrid approaches for feature extraction. For classification, support vector machines, or SVMs, were employed. The suggested hybrid method classified diseases with 98.80% accuracy.

(Nandhini and Ashokkumar 2021) presented a study that used a convolutional neural network for tomato leaf diseases detection. The dataset for this study has been taken from PlantVillage database containing 6208 images of four diseases of the leaves, including tomato mosaic virus, septoria leaf spot on tomato leaves, bacterial spot disease on leaves, and late blight. CNN models used for the classification of disease diagnosis are VGG16 and Inception V3.

The algorithm used in this study was crossover-based monarch butterfly optimization (ICRMBO) algorithm provide improvements in VGG16 and V3 models. Implementing vgg16 and Inception V3 was primarily done to improve the training process and increase CNN's classification accuracy using the ICRMBO method.

With vgg16 and inception V3, the accuracy that achieved was 99.98% and 99.94%, respectively. This proposed methodology had more sensitivity and precision and overall classification accuracy of this model was 99%.

(Sharma, Suvarna et al. 2022) presented Multi-layer convolutional neural network (CNN) for detection of mango leaves. ollection of pre-labeled leaf images and there were some changes made in model to improve the working of model. After that model was trained, and the trained model was uploaded on cloud because this was cloud based model. Python 3.4 was the software tool employed in this study. Tensorflow was the framework used in the development of the model. During data preparation step, image manipulation was done where images were resized, rescaled, and rotated in a random manner. CNN model was designed and parameter sharing was a new feature added in it. In the CNN model one layer contained 10 filters with 250 neurons, this made CNN more effective in terms of complexity and memory use. The accuracy obtained by this model was 99.27% and the validation loss was 3.88%.

(Sharma, Kaur et al. 2023) presented a data augmentation approach of generative adversarial networks to classify mango leaf diseases. This study proposed reinforcement learning algorithms for the detection of diseases. The dataset for the study was collected manually and two algorithms have been applied such as Deep Convolutional Generative Adversarial Networks (DCGAN), and Generative Adversarial Networks (GAN) model for training instances. At last all of these were compared and DCGAN considered as faster as compared to algorithms while WGAN provided good image quality.

(Kawatra 2020) proposed hybrid model using neural network for identification of diseases of mango leaves. Various algorithms in machine-learning have been developed for the purpose of detecting and classifying the mango leaf diseases, as these pathogen-caused diseases have an adverse effect on the quality of mangoes.

One study into image classification Convolutional neural network (CNN) AlexNet was considered as the best model. The suggested hybrid model perform classification using machine learning algorithm SVM and AlexNet. Several models of CNNs, such as ResNet, AlexNet, Inception V3, and VGG16 has been used for the classification of mango leaf diseases. Validation accuracy and error percentage was used for evaluation of the model. At last, comparative analysis must be conducted to choose best model among all of these models. The accuracy achieved by hybrid model is 99% that has considered as highest accuracy and outperformed other models such as including ResNet, AlexNet, Inception V3, and VGG16.

(Asha Rani K P; Gowrishankar 2023) Presented study on ConvNeXt models by detecting pathogens and pest caused illness in mango plants. In this study, deep learning was used to classify the diseased mango plants. The dataset contained data for both mango leaf diseases and mango pest dataset. The six diseases of mango leaves included in this study were infected leaves of cutting weevil, fungal disease of bacterial canker, anthracnose, powdery mildew, infected leaves with gall midge, and die back. The CONVNEXT model architecture contained ConvNeXtTiny that has 23 layers, ConvNeXtSmall that has 29 layers, ConvNeXtBase has 56 layers and ConvNeXtXLarge containing 98 and 164 layers according. The dataset of mango leaf diseases and pathogens was used to train each of these five models. All of these models have 100% training accuracy, although testing and validation accuracy can vary significantly. However, the ConvNeXtXLarge outperformed the smaller models in terms of accuracy, demonstrating that bigger models are more capable to identify complex characteristics in images. For mango leaf diseases, the accuracy was 98.8%, as for the mango pathogen dataset, the result was 100%. Both datasets achieved accuracy of 99.2% for disease classification.

2.4 Critical Analysis of Background study

The literature study of several methods to identify and classify the diseases that infected mango leaves has critically analyzed in this table, with various levels of accuracy obtained.

Table 1: Comparative Analysis of Methodologies and outcome in existing Literature

Authors	Diseases	Dataset source	Techniques used	Accuracy
(Trang, TonThat et al. 2019)	1.Anthracnose 2.powdery mildew 3. gall midge	Plant village	Convolutional neural network (CNN)	88.46%
(Sanath Rao, Swathi et al. 2021)	1. Bacterial canker 2. powdery mildew 3. scab	Self-acquired for mangoes, Plant Village	CNN	99% for grapes, 89% for mangoes
(Jain and Jaidka 2023)	1. Black soothy 2. Anthracnose	Kaggle	Hybrid approach (SVM+SGD)	97.7%
(Manoharan, Thomas et al. 2021)	1. Bacterial canker 2. powdery mildew 3. Algae leaf spot	Kaggle	CNN	98%
(Rahaman, et al. 2023)	1. Black rot 2. Scab leaf 3. Mealybug	Krishi batayon, medley, Google	DenseNet169, inceptionv3, MobilenetV2	97.81%
(P. and K. 2023)	1. Anthracnose 2. Bacterial canker 3. Die back 4. Cutting weevil 5. Sooty mold 6. Red rust 7. Gall midge	Mendley and Self-acquired	Hybrid method (EfficientNetB4+ Custom CNN)	93.01%

(Saleem, Shah et al. 2021)	1. Sooty mold 2. Powdery mildew	Self-acquired	Vein segmentation	95.5%
(S. Arivazhagan and Ligi 2018)	1. Anthracnose 2. Leaf gall 3. Leaf weaber 4. Alternaria leaf spot 5. Leaf burn	Not mentioned	CNN	96.67%
(Mehta, Kukreja et al. 2023)	Level of diseases: 1. Initial 2. Mild 3. Moderate 4. Severe	Online repositories and self-acquired	Federate learning with CNN(FL-CNN)	98%
(Faye, Diop et al. 2023)	1. Anthracnose 2. Gall midge 3. Powdery mildew 4. Sooty mold	Mendley	Data augmentation methods (affine transformation, flip, zoom, and blur)	98.54%
UDAY PRATAP SINGH et al. [12]	1. Anthracnose	Plant Village and self-acquired	Multilayer convolutional neural network (MCNN)	97.13%
Md. Rasel Mia et al. [13]	1. Shuti mold 2. Red moricha 3. Dag disease	Online repositories	Ensemble neural network NNE (SVM+NN)	80%

	4. Golmachi			
(Venkatesh, N et al. 2020)	1. Anthracnose	Plant Village	Transfer learning technique V2IncepNet	92%
(Rajpoot, Dubey et al. 2022)	Different infected leaf regions	Plant Village	Hybrid approach (BBHE and CNN)	95.65%
(Sema, Yayeh et al. 2023)	1. Anthracnose 2. Powdery mildew 3. Bacterial canker 4. Algae leaf spot	Self-acquired	Hybrid approach CNN+HOG(histogram oriented gradient)	98.80%
(Kawatra 2020)	1. Leaf rust 2. Powdery mildew 3. Bacterial spot	Plant Village	Hybrid approach (ALexNet+SVM)	99.9986%
(Asha Rani K P; Gowrishankar 2023)	1. Anthracnose 2. Bacterial canker 3. Cutting Weevil 4. Die Back	Not mentioned	ConvNeXt models	99.16654%

2.5 Conclusion

This chapter discusses current methods that are often used to diagnose and classify diseases of plant leaves. In the last several years, a significant amount of study has been conducted in the agricultural sector on disease diagnosis. These days, deep learning and machine learning models provide more accurate, dependable, and cost-effective solutions for the detection and classification of agricultural diseases.

These automated models developed for disease detection significantly perform prediction, classification and detection of diseases. In existing literature, different image preprocessing and feature extraction techniques have been discussed along with their classification algorithms. On the basis of models performance parameters, models achieved higher accuracy and less false positive results.

Chapter 3

PROBLEM STATEMENT

3.1 Introduction

Mango is one of the world's most economic fruit that plays a vital role in both the local economy and global trading. However there are several diseases on the leaves that affect the fruit yield overall. Early detection and proper remedies for the diseases can help to increase mango production and fruit quality as well. Traditional methods for disease detection were time-consuming and costly because they relied on visual inspection of human experts such as plant pathologists, which was subjective and error-prone. In recent years, advancement in technology has provided ease with applications of machine learning and deep learning techniques in agriculture in automating disease detection processes. The results of disease detection became more accurate and efficient over time.

3.2 Problem Statement

The problem addressed in this research is the development of an enhanced automated system for mango leaf disease detection using deep learning techniques. Despite there being many solutions available for the automated detection of mango leaf diseases there is always a need for improved accuracy, robustness, and efficiency in the diagnosis of leaf diseases. The specific diseases targeted in this research include Anthracnose, Bacterial canker, Powdery mildew, Dieback, Gall midge, Sooty mould, and Cutting weevil.

In this research, the problem statement revolves around the development of an advanced mango leaf disease detection system that leverages deep learning techniques to enhance the performance of existing algorithms and optimize the models parameters. The main objective of this research is to develop a solution that is capable of accurately classifying different types of mango leaf diseases while overcoming the limitations of associated individual algorithms in this research.

3.3 Problem justification

Several factors justify the development of an automated system for diagnosis of mango leaf diseases:

1. Accuracy improvement:

This research deals with a dataset that contains seven categories of diseases and works on improved accuracy, as many existing algorithms may struggle to achieve higher accuracy for the accurate identification of mango leaf diseases.

2. Computational efficiency:

The proposed deep learning techniques should be computationally efficient for the identification of diseases with accuracy or provide real-time disease detection in agriculture settings.

3. Robustness:

There are few challenges tackled in this research regarding to image dataset captured under varying conditions such as lighting, background and camera angle. All of these factors introduce noise in the dataset challenging model's robustness.

4. Technological Advancements:

Recent advancements in machine learning and deep learning algorithms, combined with high-resolution technologies of imaging provide more opportunities for the development of accurate and efficient automated disease detection systems.

3.4 Scope of study

The scope of this research will focus on the development of an automated system for mango leaf disease detection using deep learning techniques for the seven major diseases: Anthracnose, Bacterial canker, Cutting weevil,Dieback, Gall midge, sooty mould, powdery mildew. This study will utilize image data taken from the online database Kaggle that is collected from various farms or mango orchards in Bangladesh. Additionally the performance of the proposed system will be evaluated using F1, accuracy, precision, and recall.

3.5 Objectives

The primary objectives of this research are as following:

- Collecting dataset of mango leaves disease representing different diseases
- Exploring and selecting most appropriate deep learning technique for diagnosis of mango leaf diseases.

- To analyze the most effective algorithm and their parameter settings, enhance the accuracy of detection system and reduce false positive.
- Training and optimizing proposed model using collected dataset for generalization of model and achieving higher accuracy.
- To reduce the consumption of computational resources and processing time while maintaining the accuracy of the system.
- To evaluate the influence of image processing techniques on enhancing the performance of classifying mango leaf diseases.

3.6 Significance of the study

The successful development of an automated system for diagnosis and classification of mango leaf diseases has several potential benefits:

Improved crop health:

Timely detection of leaf diseases can help in preventing mango production losses and management of remedies for diseased leaves that will help in improvement of mango crops.

Cost efficiency:

An automated disease detection system reduces cost of manual labor costs that was used in traditional diagnosis system and has a better impact on cost saving for farmers.

Sustainability:

By implementing targeted and precise automated system for disease diagnosis and classification contribute to sustainable agriculture practices and in environmental stewardship.

In conclusion, development of an automated system for disease diagnosis and classification of mango leaves address a critical exigency in agriculture research and has revolutionize disease management practices in mango cultivation. This research aims to contribute to the advancement in precision agriculture and harnessing the power of deep learning and image processing techniques.

Chapter 4

PROPOSED METHODOLOGY

4.1 Introduction to methodology

This chapter aims to present suggested methods and deep learning algorithms used for detection of mango leaf diseases. The proposed methodology has divided into 4 steps that incudes data collection, data preprocessing, splitting dataset into training and testing, model development and at last performance evaluation. Data preprocessing helps in data uniformity and made dataset more balanced. After preprocessing, dataset has been divided into training and testing data. Data has been separated by randomly dividing dataset into two parts: training set and testing set. These sets contain eighty percent images and twenty percent images. The deep learning algorithm customized convolutional neural network (CNN), AlexNet, and squeezeNet models are presented in this research for identification of mango leaf diseases. This chapter includes detail discussion of preprocessing techniques and proposed models.

Proposed techniques for classification of mango leaf diseases are customized convolutional neural network (CNN), Alex Net and Squeeze Net and RMsprop training optimization algorithm is used for training the models. CNN is most widely used algorithm for image classification in Deep learning.

Below figure shows the proposed methodology for this research.

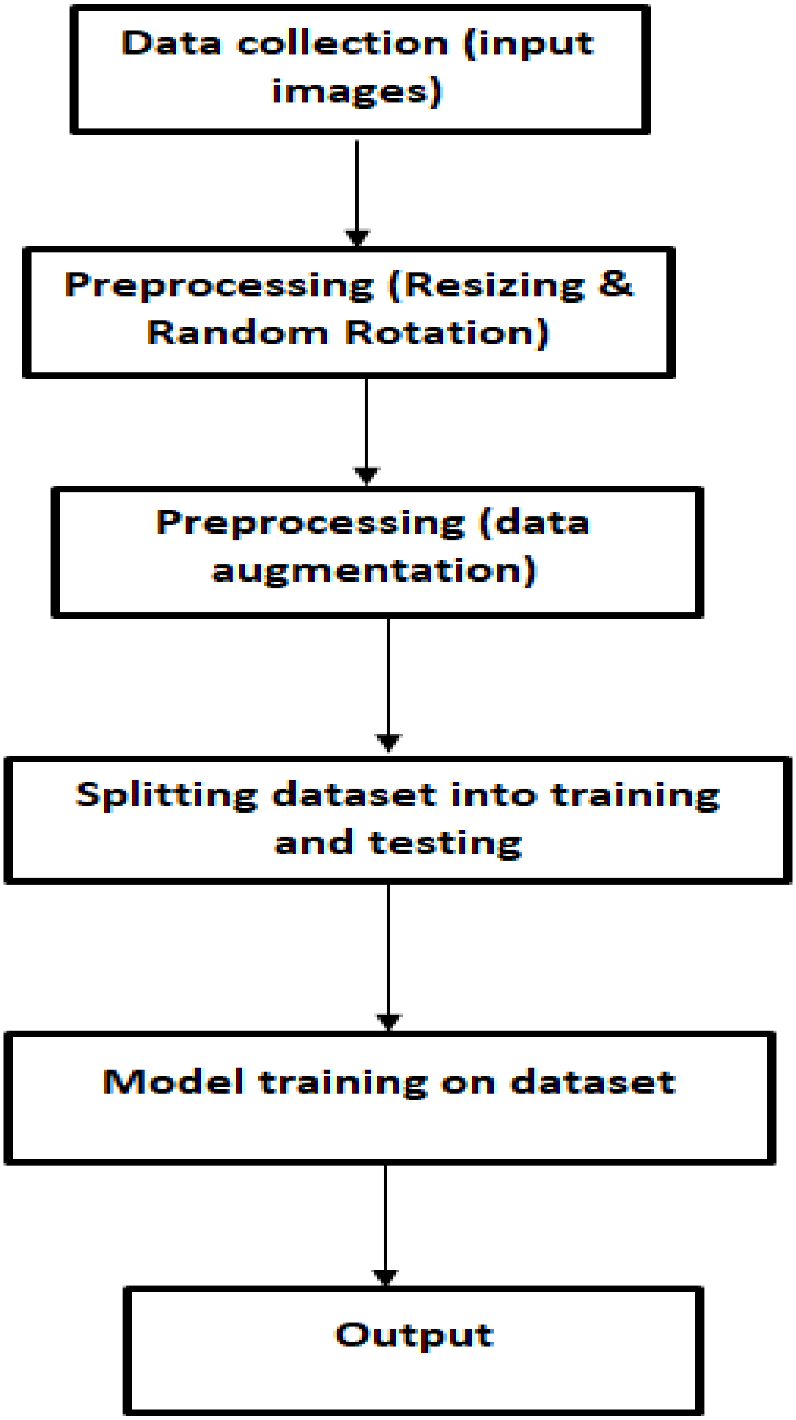

Figure 5: Proposed Methodology

In this figure, first step is data collection followed by data preprocessing that involved image resizing and data augmentation. Data augmentation increase model robustness and mitigate overfitting. The dataset has to be trained on each of the three models—CNN, Alex Net, and Squeeze Net—before being divided into training and testing datasets.

4.2 Data collection and preprocessing

The approach for developing deep learning automated system for the detection of mango leaf diseases is described in this part. There are three necessary components for developing an automated system is data collection, data preprocessing, and implementation of Convolutional neural network (CNN). The proposed CNN model must be trained and tested on large number of images of mango leaves diseases.

4.2.1 Data collection

To get start with this work, dataset containing large number of images has collected. There are several ways for data collection including field trips, internet database such as Kaggle and different agriculture research facilities.

In this research dataset has been acquired from online source (Kaggle) that provides a wide variety of datasets on mango leaves diseases. The obtained dataset includes 4000 images of 240x320 pixel representing various leaf disease categories such as infected leaves of anthracnose, diseases bacterial canker, cutting weevil diseases on mango leaves, gall midge, fungal diseases on leaves like powdery mildew, die back, sooty mold diseases, and a healthy images of leaves. This dataset was collected from mango orchards at, University of Jahangir Nagar and Sher-e-Bangla Agricultural, Village of Udaypur, and Itakhola Village in several Bangladeshi farms. It is considered that deep learning algorithms are hungry algorithms because they require a large dataset for proper results (Ahmed, Ibrahim et al. 2023).

4.2.2 Data preprocessing

The dataset undergoes many transformations throughout the data preparation stage in order to increase the dataset's quality and provide better model performance.

Data preprocessing step carried out as data preparation stages that improve data quality and data consistency. There are few steps included in this research for data preparation including:

- Resizing
- Data augmentation

4.2.2.1 Resizing

The dataset resized to a standard dimension to ensure uniformity across the dataset such as 227×227. Resizing helps in reducing the computational complexity of models. Some models have input requirement to resize images. For this research, 4000 images dataset that were of different resolutions, were resized as 227x227 pixels in each class and make sure that model can efficiently process the images. The image dataset consisted of three channels (Red, Green, and Blue) in RGB format.

4.2.2.2 Data augmentation

By applying several modifications to the dataset, the data augmentation approach is primarily employed in the field of deep learning to increase the size of training dataset artificially. Common techniques used in data augmentation include flipping, rotating, cropping, scaling, and shearing.

Data augmentation technique considered as most powerful method to made models more robust and generalized. In this research, the following augmentation techniques have applied on dataset:

- Random rotation
- Random reflection
- Random X shear
- Random Y shear

Shearing is data augmentation technique applied on dataset to generate variation in original images. Shearing is extremely helpful technique for stimulating different angles at which images of the leaves might be captured. Random rotation parameter rotates the images in clockwise or ant clockwise direction within the range of 5 to -5 degrees in this study. Random reflection is also known as horizontal flip and it control horizontal reflection on images. Random X shear and Y shear applied on dataset in vertical and horizontal direction within the range of 0.05 to -0.05 in this study. The shearing effect distorts the image by shifting pixels in vertical and horizontal direction.

4.3 Model Training and testing methodology

In this approach, dataset of the affected images trained on 80% of total data. For train and test split, dataset has been divided into 80% and 20% of images. The dataset has divided into two parts including training and testing sets as basic requirement for evaluating the model performance. After preprocessing of the acquired images, CNN models used some layers to extract the features and perform classification on these images later for mango leaf disease detection.

The dataset split in this research is given below. This table shows diseases and number of training images in each disease class. Each disease class contain 400 training images.

Table 2: Training Dataset

Disease	Training images
Anthracnose	400
Bacterial canker	400
Cutting weevil	400
Die back	400
Gall midge	400
Powdery mildew	400
Sooty mould	400
Healthy	400

In this split, dataset contain testing images of mango leaf diseases. The model's

weight is updated using training dataset,and testing set provides an insight to tune the hyper parameters and monitor the performance during training.

Table 3: Testing Dataset

Disease	Testing images
Anthracnose	100
Bacterial canker	100
Cutting weevil	100
Die back	100
Gall midge	100
Powdery mildew	100
Sooty mould	100
Healthy	100

The testing dataset is used to evaluate the performance of model by calculating the F1score, recall and accuracy.

4.4 Model development

The latest developments in deep learning models used in this research are the primary focus of this section. Convolutional neural networks (CNNs), AlexNets, and SqueezeNet architectures are the first, second, and third models, respectively. These CNN models are used in this research for classification of mango leaf diseases.

4.4.1 Convolution Neural Network (CNN)

Convolutional neural network (CNN) is considered as most widely used deep learning model for classification of images and visualization. The CNN has been developed specifically for structured data sets, such as image data. For identifying objects, image segmentation, and image recognition, convolutional neural networks (CNNs) are employed.

Convolutional neural networks (CNNs) are used to resolve issues with image processing. Projects involving natural language processing can also make advantage of CNN. One of the key reasons CNN is so crucial in the field of deep learning, machine learning, and in artificial intelligence because it help these disciplines to grow quickly. In contrast to other neural networks, CNN is considered to be efficient in processing and classification of images. These three layers make up a neural network: the input, hidden, and output layers.

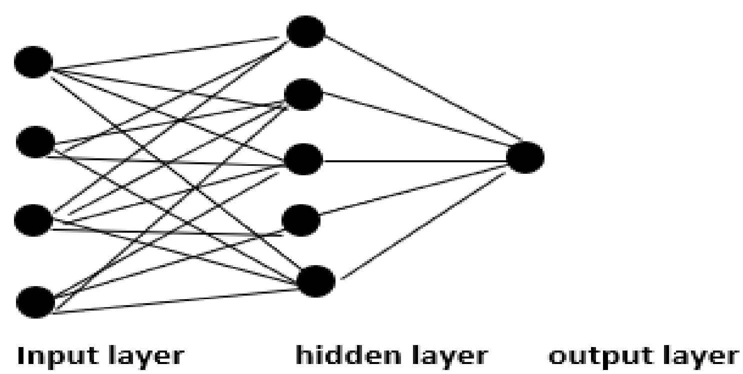

Figure 6: Neural Network Layers

The input layer accepts different types of input. The hidden layer calculates these inputs. After that, the output layer provides the computation and extraction results. Every neuron in these levels has a weight of its own and is coupled to the neurons in the layers above it.

When handling images or languages, the convolutional neural network (CNN) operates differently since it treats the data as spatial. Then, neurons simply connected to the neurons adjacent and all neurons have the same weight, rather than neurons being connected to every neuron in the preceding layer.

The network maintains the geographical component of the dataset because of its connection simplification. The process of filtering that occurs during this operation is referred known as "convolutional." For instance, CNN reduces complicated images to make them easier to understand and interpret.

4.4.2 CNN Architecture

Convolution Neural Networks (CNNs) are composed of many layers, much like other neural networks. Following are the layers of CNN:

- Convolutional layers
- Pooling layer
- ReLu Layer
- Fully connected layer

4.4.3 ReLu layer

By serving as an activation function, this layer makes sure that data flow nonlinearly across the network's layers. Consequently, the convolutional layer was employed before this layer. Without it, the dimensionality that we need to keep would be lost as data was put into each layer. Calculations may be performed on the provided dataset according to the fully connected layer.

4.4.4 Convolutional layer

Convolutional layer is the most crucial layer. The feature map of convolved layer is created by implementing a filter on a range of image pixels. This layer acts like a window to see all the features that would not be visible otherwise.

4.4.5 Pooling layer

CNNs usually use two techniques for pooling. The first technique, known as local pooling, aggregated data from tiny sections, such 3 x 3, to create a feature map. Global pooling is consider as second technique which assign scalar value to each feature on feature map that represents the feature vector of images. This depiction is derived from fully connected layers. Pooling layer provide an advantage by reducing spatial dimensions and time complexity and made a huge impact on model over fitting risk.

By pooling, the sample size of a given feature map can be reduced or scaled down. Because it decreases the amount of parameters the network must analyze, this layer also speeds up processing significantly. This layer produces a pooled feature map as its output. The CNN model has two methods for pooling:

1. Max pooling: This method utilizes a given convolved feature's maximum input. 2. Average pooling: This method only calculates the convolved feature average. By using its own set of mathematical principles, the network builds up a picture of the image data through these procedures in the pooling layer, which amount to feature extraction.

4.4.6 Fully connected layer

This layer is utilized for dataset classification. This layer's neurons are all linked to the layer above it, resulting in the easier classification of the whole feature set that had been previously learned. The output of this layer is a vector of N-dimensions, where N represent the total number of classes. Only linear data can be processed by a neural network with more intricate connections. Fully connected layer uses the activation feature of softmax for grouping accurately and generating possibilities from zero to one (Manoharan, Thomas et al. 2021).

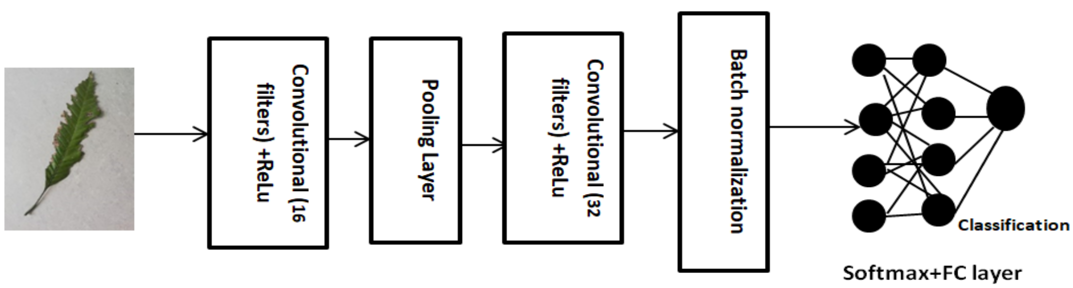

Figure 7: CNN Architecture

4.4.7 Advantage of CNN

With advancement in technology, deep learning networks have grown immense for image classification, pattern recognition, object detection, and voice recognition. The concept of deep learning has been motivated due to convolutional neural networks. CNN can handle large and complex data like image effectively. CNN can learn the hierarchical representation and allow automatic feature extraction, (mention layer that extract features).

The most effective reason for the popularity of CNN in neural network us that it is adaptable to various tasks with minimal manual feature engineering.

4.4.8 Disadvantages of CNN

Deep learning models like CNN and all other neural models require large amount of data and they work well with larger datasets. In CNN model, over fitting can occur if model is not properly regularized or the dataset is too small.

4.4.9 Applications of CNN in Deep learning and disease detection

There are many applications available of deep learning provide using CNN:

- Deep learning revolutionized many fields especially in computer vision where CNN is making huge progress due to its ability of hierarchical feature learning automatically.
- In past few years, deep learning has been widely used in agriculture and making efforts for developing, controlling and in enhancing the agriculture production (Singh, Chouhan et al. 2019).
- CNN is considered as best tool of deep learning that process large dataset with large number of classes provide the facility of pattern recognition in image processing applications (Kumar, Ashtekar et al. 2021).
- Deep Learning helps CNN several advantages like automatic feature extraction, scalability to larger dataset and robustness to variation in data.

4.5 Alex Net

The deep convolution neural networks (CNN) architecture known as Alexnet was created by Geoffrey Hinton, Ilya Sutskever, and Alex Krizhevsky. In 2012, this design became widely known after winning the large-scale ImageNet visual recognition competition. AlexNet is a CNN design with eight layers that is deep. Three completely linked layers and five CNN layers make up the AlexNet architecture. Information is extracted from the input images using the first five segments of the CNN model. Filters in these levels might differ in size and quantity. This architecture makes use of dropout regularization, local response normalization, and the ReLu activation function.

The main reason for the popularity of AlexNet that it has ability to extract high level features from the images making it appropriate for complex classification tasks like disease diagnosis and classification. In this research, we choose Alexnet model because it captures complex image patterns with simplicity.

Architecture of AlexNet is given below:

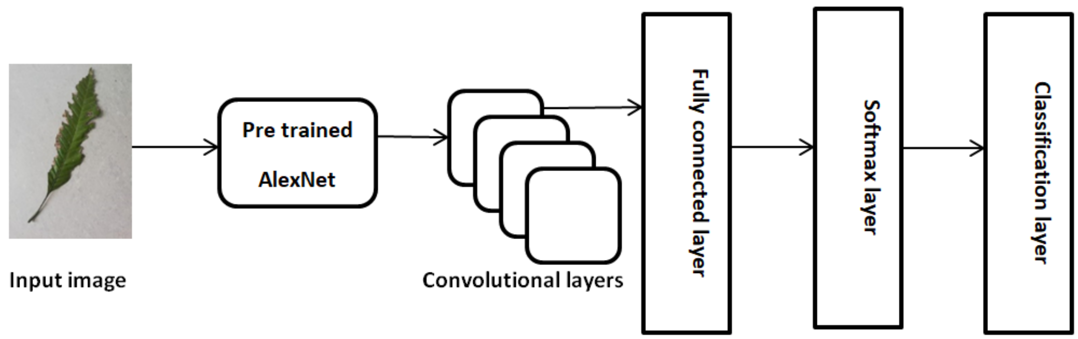

Figure 8: Alex Net Architecture

This diagram depict architecture of Alex Net which consist of five convolution layers used for feature extraction and subsequently, there is the classification layer, softmax layer, and the fully connected layer.

4.5.1 Advantages of Alexnet

Here are some advantages given for choosing AlexNet for this study:

- AlexNet model provide a deeper insight into architecture and extract more complex features from data as compared to other models.
- AlexNet model demonstrate amazing performance in image classification tasks by learning the depth of hierarchical representation of features of images.
- AlexNet model introduce dropout regularization during training. This dropout randomly disables some neurons during training, and force model to learn more robust and generalized features of the images.

4.6 Squeeze Net

SqueezeNet is well known model for its lightweight architecture. This model can achieve highest accuracy with fewer parameters. It reduces large filters with 1 ×1 filters to reduce model size and computational complexity and making it more efficient. SqueezeNet is widely used in crop disease detection because it is compact model than other CNN models such as AlexNet. squeezeNet architecture consist of three parts: one is filter size reduction, second part reduce the input channel, and third part is the dawn sampling of network's conclusion. SqueezeNet model make use of fire module that is predefined within architecture. By making call of squeezeNet, it automatically makes use of fire module, pooling layers and final classification layer. Architecture of SqueezeNet is given below:

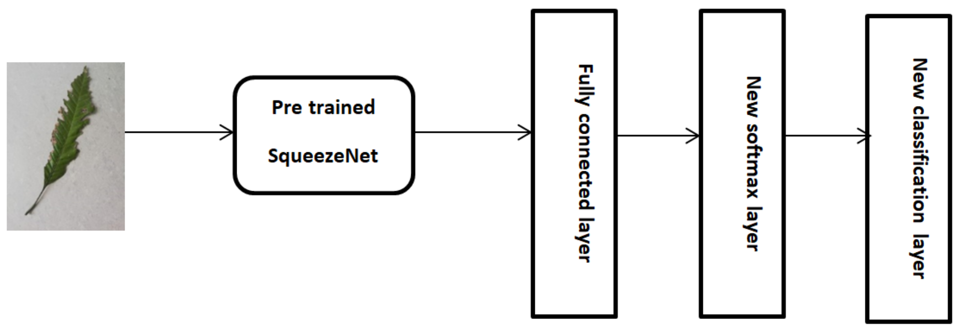

Figure 9: Squeeze Net Architecture

In the above figure, squeezeNet model has loaded and then last three layers which are replaced with new layers. There is one last pooling layer (pool 10) which is replaced with fully connected layer, there is an addition of new softmax layer in place of the previous layer of softmax (probability), and a new classification layer is added in place of the last one.

4.6.1 Working of SqueezeNet

SqueezeNet model layers breakdown into following:

- Input
- Convolutional layers
    1. Convolutional layers(ReLu activation, max pooling)

2. Fire module (1-8)

- Squeeze (1×1 convolutional, 16 filters)
- Expand (1×1 convolutional, 64 filters), 3×3 convolution(64 filters)

3. Fully connected layer
4. New softmax layer
5. New classification layer

There are total 8 fire modules in squeezeNet architecture and number of filters increase gradually as per fire module. In this architecture, squeeze layer and expand layer are the most important part. Squeeze layer reduces the size of input channel from 3×3 to 1×1. Basically, the squeeze layer responsible for guide expand layer to reduce the input channel by using fire module. The model turned out to be very simple because expand layer receive less parameters. Layers 1, 4, and 8 are utilized before the max pooling layer with stride 2 (Priya Ujawe 2023). Last layers are the fully connected, two replaced layers in this architecture because of custom output requirement.

Advantages

- squeezeNet is simple and lightweight model as compared to other models like AlexNet.
- It have small architecture that uses less bandwidth(Priya Ujawe 2023).

4.7 Performance evaluation

Performance metrics, confusion matrix and cross-validation have used in this research for evaluation of CNN models performance. Here's detail discussion of confusion matrix given below:

4.7.1 Confusion matrix

A table termed a confusion matrix is used for evaluating how effectively machine learning and deep learning models perform. This matrix is famous as confusion matrix because model it display the instances where model is confused and made mistakes.

Table 4: Confusion Matrix

Confusion matrix		
	Actual Negative	**Actual Positive**
Predicted Negative	True Negative (TN)	False negative(FN)
Predicted positive	False Positive (FP)	True Positive(TP)

This table helps model to predict the True positive, True Negative, False positive, and False Negative classes. In True positive case, model predicts correctly True positives, while on the other hand in False Negative all the prediction for the classes are incorrect. The True negative label can predict all negative classes correctly while, the False positive made prediction for negative classes incorrectly.

Confusion matrix basically analyzes the model performance for each class such as diseased class and healthy class and provides an insight to identify the misclassification. The F1-score, recall, accuracy, and precision are among the additional metrics that are computed using the confusion matrix. Each of these indicators is useful for determining the way the model performs.

4.7.2 Metrics for classification performance

In order to evaluate the performance, we need to calculate the Accuracy, precision, recall, and F1-score of each model.

4.7.3 Accuracy

While measuring performance, accuracy can be used to calculate the percentage of accurately predicted predictions out of all the predictions made. It is used to measure that how often model give correct results.

$$\text{Accuracy} = \frac{\mathbf{TP+TN}}{\mathbf{TP+TN+FP+FN}}$$

4.7.4 Precision

Precision is used to compute the proportion of predicted positives that how many are actually positives. It is applied to evaluate the models accuracy in predicting positive instances.

Precision= $\frac{\mathbf{TP}}{\mathbf{TP+FP}}$

4.7.5 Recall

Recall is used to compute the proportion of true positive instances. Recall is used to predict the positive instances of a model.

Recall=$\frac{\mathbf{TP}}{\mathbf{TP+FN}}$

4.7.6 F1-Score

The accuracy and recall average are calculated using the F1-score. It offers a metric that achieves an ideal balance between accuracy and recall.

F1-score= $2\times\frac{\mathbf{Precision\times Recall}}{\mathbf{Precision+Recall}}$

Cross-fold validation

Cross-validation performed in this research and the dataset has divided into 5 cross k-fold that obtain more robust estimates of model’s performance.

4.8 Training optimizer

RMSProp (Root Mean Square Propagation) is most popular training optimizer used in deep learning. It is an enhancement of the recognized stochastic gradient descent (SGD) technique, which was developed in order to address the issue of deep learning models' reducing gradients. Based on the gradient's magnitude, this optimizer adjusts the learning rate of each parameter separately.

Adaptive learning rate, which modifies the learning rate based on the gradient's magnitude for every parameter and square root the learning rate to maintain the learning process, is one of

RMSProp's key characteristics. Three deep learning CNN models including AlexNet, squeezeNet trained on RMSProp Algorithm in this research.

4.9 Software tool

MATLAB software has used for classification of mango leaf diseases. In MATLAB, editor window is used to write the run code. This software contains deep learning toolbox that made our task easier to train dataset.

Chapter 5

EXPERIMENTS AND RESULTS

5.1 Introduction

In our experiment there are total eight classes available in dataset, seven classes are of diseased category including Anthracnose, Bacterial canker, Cutting weevil, Die back, Gall midge, Powdery mildew, Sooty mould.

Three Algorithms are applied on this dataset. First one is Squeeze Net, second is Alex Net and third is CNN . The 5 fold cross validation test are used for the classification measurement.

5.2 Squeeze Net model classification result

The given table presents accuracy, precision, recall, and F1-score of the squeeze Net model. In this research, squeeze Net model achieved highest accuracy of 99.6% and perform better as compared to other two models.

Table 5: Results of Squeeze Net Model

Squeeze Net model for diagnosis and classification of mango leaf diseases	
Accuracy	99.6%
Precision	99.8%
Recall	99.4%
F1-score	98.4%

Confusion matrix of Squeeze Net:

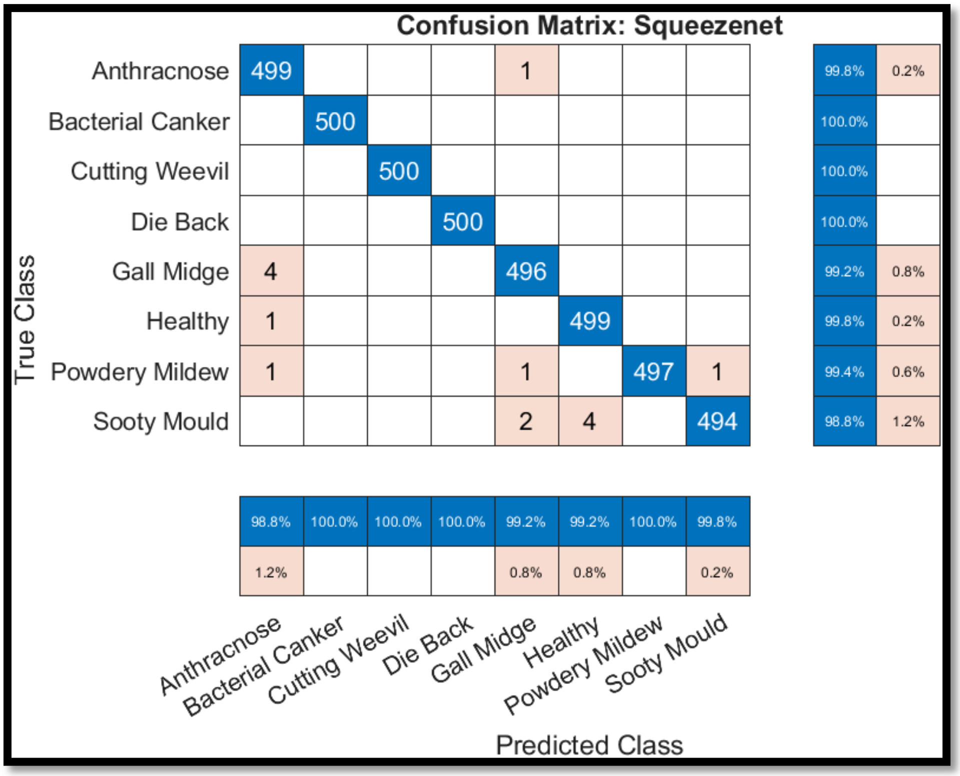

Figure 10: Confusion matrix Squeeze Net model

Explanation of matrix:

The confusion matrix in the above figure display the predicted class labels and true class labels of Squeeze Net model and show the row summary and column summary of the result. True classes of Anthracnose are 499 and 1 is misclassified. True classes of Anthracnose are 99.8% correctly classified and 0.2% misclassification and predicted class shows 98.8% true classes correct classified and 1.2% misclassification. Bacterial canker, cutting weevil, die back has 100% true classes and no misclassification found in these diseases. Gall midge class has 99.2% in true class and 0.8% misclassification in true class. Same percentage for gall midge have found in predicted class. True class of healthy category has 99.8% and 0.2% of the misclassification found. Powdery mildew has true class 99.4% and 0.6% misclassification found in true class while on predicted class it has 100% classification. Sooty mould has true class of 98.8% of correctly classified while 1.2% instances are misclassified. Predicted class has 99.8% correct classification and 0.2% of misclassification found.

5.3 Alex Net model classification result

AlexNet brings this result when this model has applied on mango leaf disease dataset. This table shows accuracy, precision, recall, and f1-score of the dataset.

Table 6: Results of Alex Net Model

AlexNet model for diagnosis and classification of mango leaf diseases	
Accuracy	92.5%
Precision	96.12%
Recall	98%
F1-score	95.2%

Confusion matrix of AlexNet:

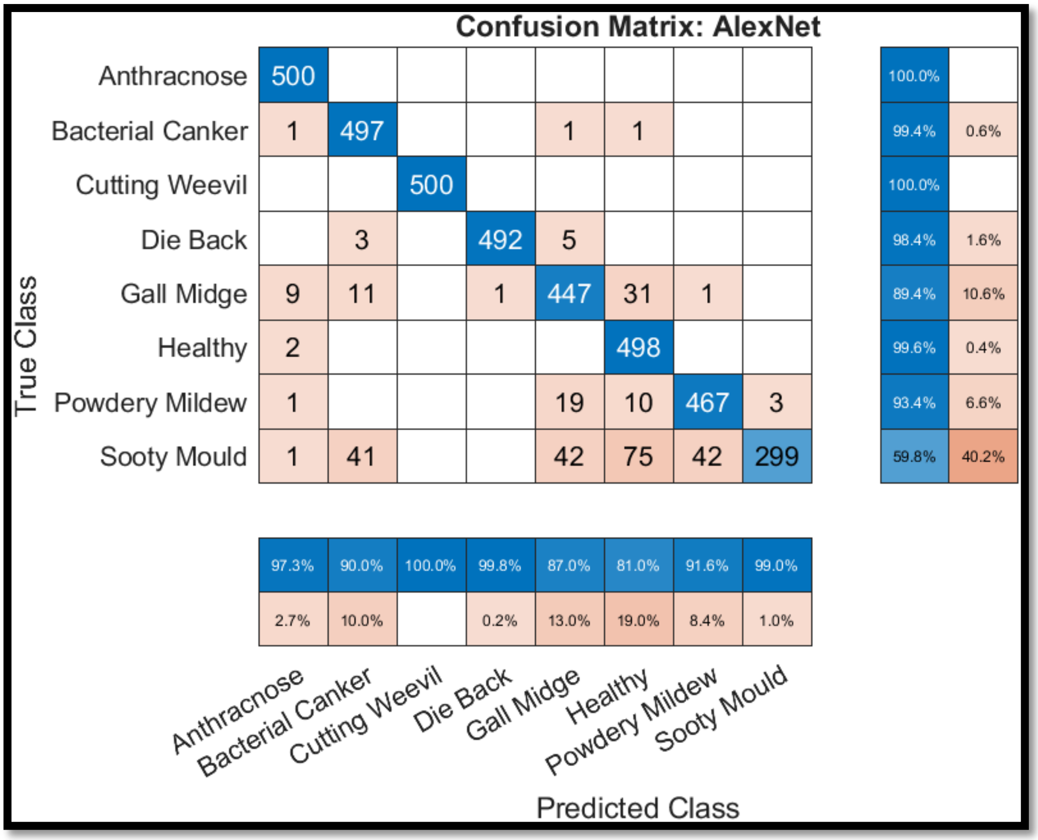

Figure 11: Confusion matrix Alex Net model

Explanation of matrix:

The confusion matrix in the above figure display the predicted class labels and true class labels of Alex Net model. The class of Anthracnose shows that all 500 instances are correctly classification. True classes of Anthracnose are 100% and predicted class shows 97.3% true classes are correctly classified and 2.7% misclassification. Bacterial canker has correctly classified 99.4% and 0.6% misclassification in true class while in predicted class it shows 90% correctly classified results and 10% misclassification. Cutting weevil has 100% true classes and no misclassification found in this diseases. Gall midge class has 89.4% in true class and 10.6% misclassification in true class, while on the other hand predicted class has 87% correctly classified instances and13 % misclassified instances. Healthy category of images has 99.6% of correct classification in true class while 0.4% of misclassification and in predicted class has 81%

correct classification and 19% of instances has misclassified. Powdery mildew has 93.4% of correct classification in true class while on the other hand 91.6% correct classification in predicted class. Sooty mould has 99% of correctly classified instances in predicted class and 59.8% of correctly classified in true class. Only sooty mould shows lowest correct classification in this matrix with 40.2% of misclassification in true class.

5.4 CNN classification results

This table shows CNN model classification results including accuracy, precision, recall and f1-score.

Table 7: Results of CNN model

Table 3: CNN model for diagnosis and classification of mango leaf diseases	
Accuracy	93.75%
Precision	98.9%
Recall	98.95%
F1-score	96.47%

Confusion matrix of CNN:

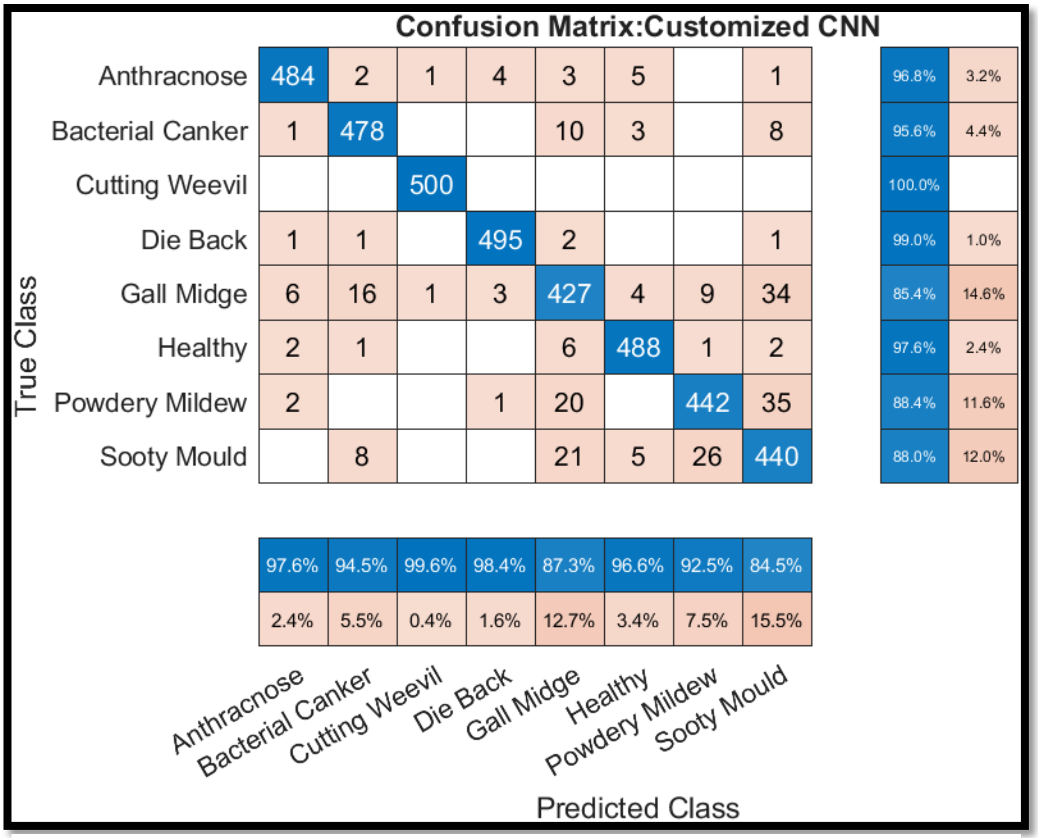

Figure 12: Confusion matrix Customized CNN

Explanation of confusion matrix:

The confusion matrix in the above figure display the predicted class labels and true class labels of CNN model and show the row summary and column summary of the result. True classes of Anthracnose are 484 correctly classified instances and 16 instances are misclassified. True classes of Anthracnose are 96.8% correctly classified and 3.2% misclassification and predicted class shows 97.6% true classes correct classified and 2.4% misclassification.

Bacterial canker has 95.6% correct classification in true class and 94.5% correct classification in predicted class. Bacterial canker has 4.4 % misclassification in true class and 5.5% of misclassification in predicted class. Cutting weevil has 100% correctly classified instances in true class and 99.6% correct classification in predicted class.

Die back has 99% of correctly classified instances in true class and 98.4% in predicted class. So this shows a slight variation between correct classifications. Gall midge has 85.4% correct classification in true class while 87.3% in predicted class. Healthy category has 97.6% of correct classification in true class and 96.7% in predicted class. Powdery mildew has 92.5% sooty mould has 84.5% correct classification in predicted class while in true class these two have 88.4% and 88% of correctly classified instances.

Chapter 6

COMPARISON OF MODELS

This chapter based on the comparative analysis of three CNN models implemented in this research. Following table shows the accuracy, precision, recall, and F1-score.

6.1 Comparison of CNN models

Table 8: Comparison of Models

Results	SqueezeNet	AlexNet	CNN
Accuracy	99.6%	92.5%	93.75%
precision	99.8%	96.12%	98.5%
Recall	99.4%	98%	98.95%
F1-score	98.4%	95.2%	96.47%

In mango leaf disease detection, SqueezeNet model achieved highest accuracy that is 99.6% followed by AlexNet model which achieved 92.5% of accuracy and CNN achieved 93.75% of accuracy in disease diagnosis and classification. All of three CNN models perform well with image classification.

6.2 Comparison of model on basis of correct classification of three models

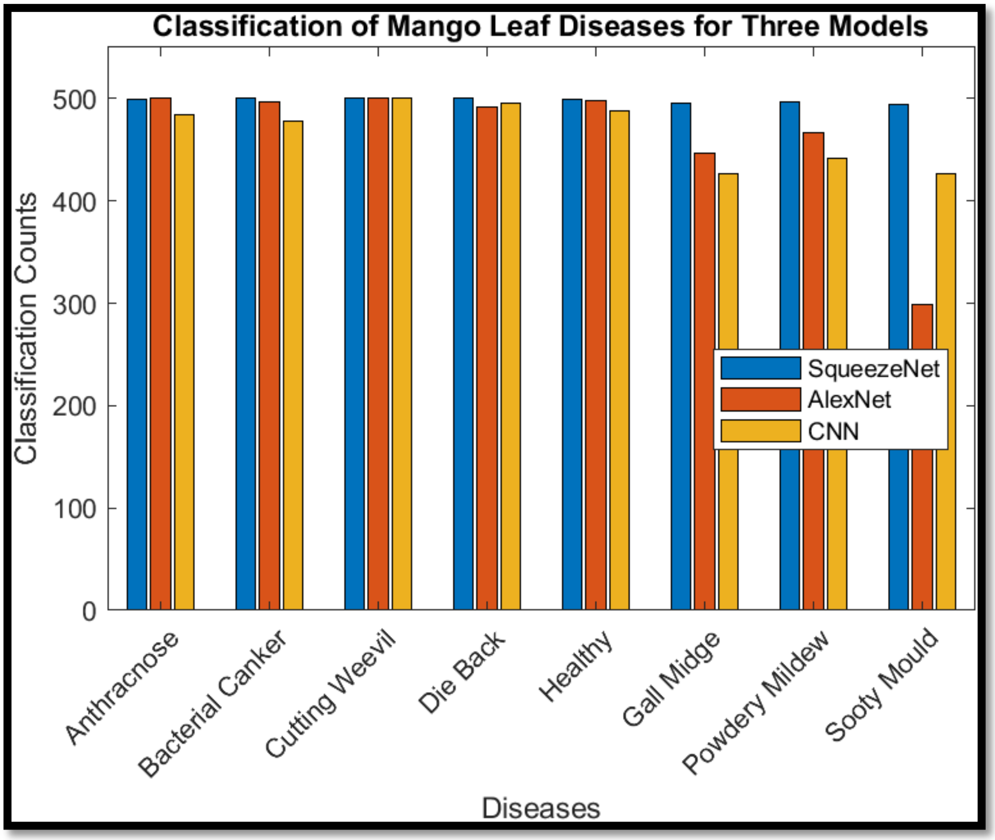

Figure 13: classification of models

This graph shows the classification of dataset on three models and squeezeNet model has high rate of classification achieved in this research.

Visualization of misclassification of models:

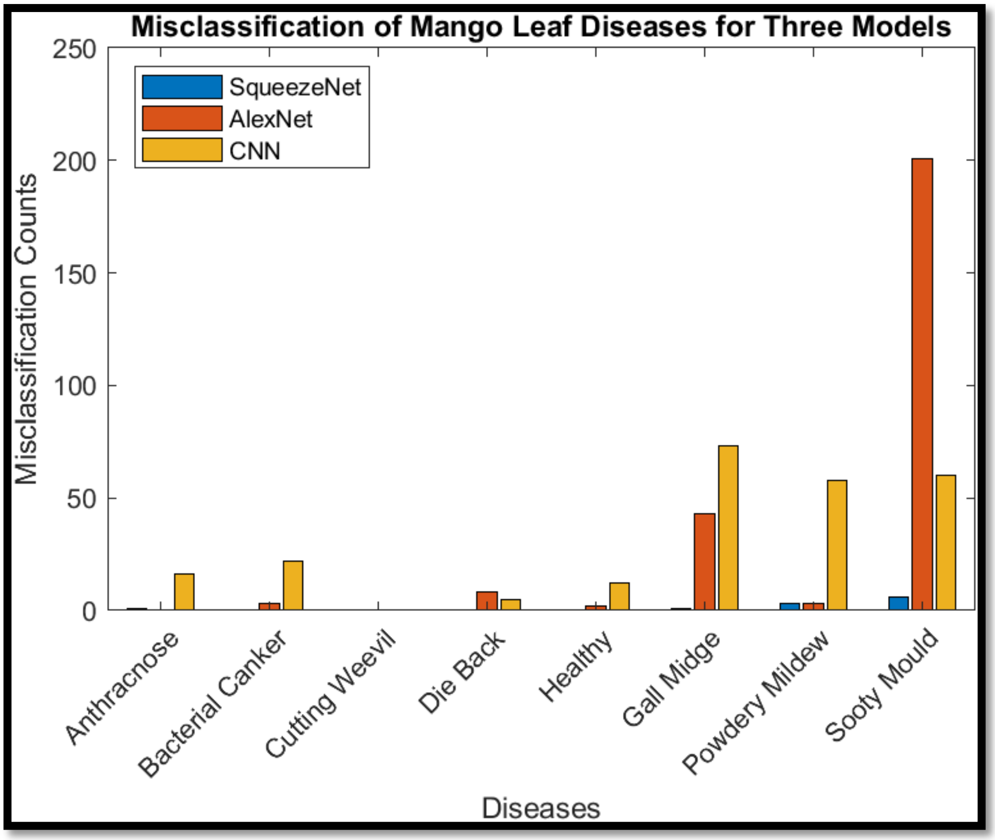

Figure 14: misclassification in models

This is the graph of all of the three models misclassification rate in diagnosis and classification of mango leaf diseases discussed in this research.

Chapter 7

CONCLUSION AND FUTURE DIRECTION

In agriculture sector, detection of plant diseases has been a major challenge. With the advent of technology, machine learning and deep learning played significant role in detection of diseases with better accuracy. Deep learning presents different automatic models for detection of diseases. There are many pre trained deep learning models available for diagnosis and classification of diseases. In this research, three CNN deep learning models AlexNet, SqueezeNet and CNN has presented.

Dataset used in this research has acquired from Kaggle which contain 4000 images of mango leaves diseases such as Anthracnose, Bacterial canker, Cutting weevil, Die back, Gall midge, Powdery mildew, and sooty mould. Each category of the disease has total 500 images so dataset is balanced in this research.

The key aim of this study is to develop an automated deep learning system for detection of mango leaf diseases and to improve the accuracy, precision, recall of deep learning models. To achieve this objective, we have presented three CNN models including CNN, AlexNet, and SqueezeNet model. Out of three CNN models, SqueezeNet achieved better accuracy 99.6% and AlexNet achieved 92.3% and CNN achieved 93.75% of accuracy. Both of the models Alexnet and CNN achieved high accuracy but SqueezeNet outperformed in terms of accuracy and efficiency for classification of diseases. All of these models work well with given CNN models. The classification accuracy has improved more because deep learning models work well with larger datasets.

This research has significant impact on agriculture for the development of automated deep-learning based system for diagnosis and classification of mango leaf diseases.

Future work will include expansion of dataset into more disease categories of mango leaves and then training on variety of transfer learning models because they provide more robustness in disease detection. In this work, all work has performed on single CPU. In future, performance parameter will be considered and improved by implementing fast computing devices like GPUs.

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